

Fischer–Tropsch Synthesis: Effect of Potassium on Activity and Selectivity for Oxide and Carbide Fe Catalysts

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Abstract The effect of potassium on oxides and carbides of iron for Fischer–Tropsch synthesis (FTS) was investigated by pretreating Fe_3O_4 and K-promoted Fe catalysts with different gases ($\text{H}_2/\text{H}_2\text{O}$ and CO). A freshly activated sample and catalysts that were recovered from the CSTR before, during and after FT synthesis were characterized *ex situ* using Mössbauer spectroscopy. Iron carbide is found to be active for both FT and water gas shift (WGS) reactions. After $\text{H}_2/\text{H}_2\text{O}$ activation, all three catalysts (Fe_3O_4 , low α -Fe, and high α -Fe) exhibit a steady but low FT activity for a period of FT synthesis. However, both FT and WGS activity for Fe_3O_4 and low α -Fe catalysts were greatly improved after CO activation. In contrast, the high potassium containing catalyst (high α -Fe) did not show any further improvement in activity after CO activation. The difference in FT and WGS activity observed after pretreatment conditions using these catalysts may be associated to the amount of potassium and conversely the iron carbide present in the catalysts.

Keywords Fischer–Tropsch synthesis · Magnetite · Iron carbide · Potassium · Oxygenates

1 Introduction

In Fischer–Tropsch synthesis (FTS), syngas ($\text{H}_2 + \text{CO}$) is catalytically converted through a surface polymerization reaction into a wide spectrum of hydrocarbon chains [1, 2]. Iron, cobalt, nickel and ruthenium are catalytically active for FTS and among these, iron and cobalt are used in industry. Iron based catalysts have an added advantage of possessing water-gas shift (WGS) activity ($\text{CO} + \text{H}_2\text{O} \leftrightarrow \text{CO}_2 + \text{H}_2$). This makes them suitable for the production of liquid fuels from syngas derived from coal and biomass, which have typically low H_2/CO ratios that are in the range of 0.5–1.5 [3]. A typical iron catalyst often contains promoters like copper and potassium along with some silica or alumina. Depending on the activation conditions, iron may exist in a metallic, carbidic and/or oxidic form during FTS [4].

In the past, many studies have been devoted to defining the nature of iron phases present after activation and during FT synthesis [4]. However, the exact nature of active phase(s) for FT and WGS reactions remains controversial. For example, carburized iron is an active catalyst for FTS but many consider the magnetite (Fe_3O_4) phase to be responsible for WGS reaction. Some reported that Fe_3O_4 also may be an active phase for FTS [5].

Modification of Fe-FT catalysts by adding alkali promoters such as potassium has been studied for many years. It has been shown that potassium exhibits a positive [6, 7] or negative [8] effect on CO conversion and in some cases, the activity appeared to reach a maximum with increasing potassium loading and then decline with further addition of potassium [6]. For instance, Raje et al. [9] found that potassium containing Fe catalysts exhibit an increasing dependency of the FTS on the hydrogen formed by WGS reaction. Potassium was also found to influence the product selectivity of Fe-FT synthesis; in particular, it was known

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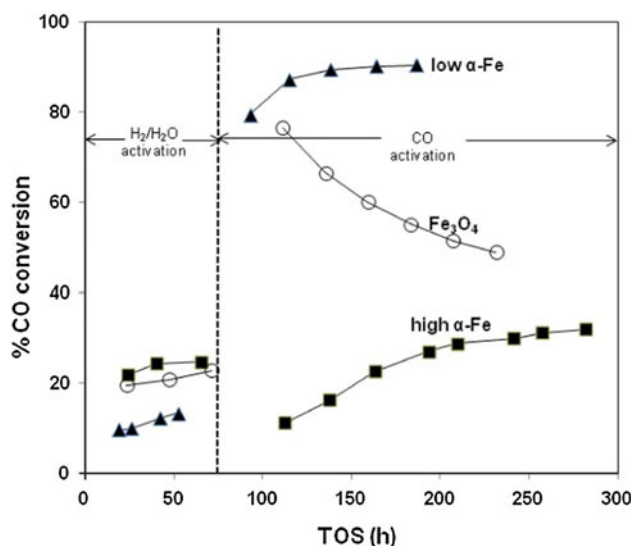


Fig. 1 Change of CO conversion with time on stream for unpromoted and Cu, K-promoted Fe catalysts after different pretreatment conditions. [CO activation condition: CO (3 sl/h/g cat.), 543 K, 1.3 MPa, 24 h; $\text{H}_2/\text{H}_2\text{O}$ activation condition: 513 K, 1.3 MPa, $\text{H}_2/\text{H}_2\text{O} = 1.0$, 24 h; FTS condition: 543 K, 1.3 MPa, $\text{H}_2/\text{CO} = 1.0$, $\text{SV}_{(\text{syngas})} = 3.0$ sl/h/g cat.] (broken line is where CO activation occurred)

to increase wax and alkene yields while decreasing the production of undesirable methane [10]. Several factors influence the activity and selectivity of potassium promoted iron catalysts and among these, potassium presumably increases CO dissociation as one of the factors that determine FT and WGS activity. In other words, potassium promotes FT and WGS activity of iron by keeping them in an active carbide phase [4].

In spite of the recognized importance of the promoter K in Fe-FT synthesis, data to document their interactions with iron carbides and iron oxide (magnetite) are not abundant. Sorescu et al. [11] proposed, based on DFT calculations, that K decreases the interaction energy of CO by stabilizing C and O species in the transition state, with a net favored interaction between K and O. The authors further claimed that this change of transition state geometry is due to charge donation leading to increased separation of C and O atoms from the surface and reduction of the bonding competition. Van Steen and Claeys [12] recently demonstrated the state and interaction of K with χ - Fe_3C_2 carbide using molecular modeling. The authors argued that the enhancement in CO conversion and the changes in FT selectivity probably originates from the electronic interactions between K and the surface of Fe. However, under realistic FT synthesis conditions in addition to carbides (χ , ϵ , and θ), some oxides of Fe (Fe_3O_4) are present and thus it is necessary to understand their individual interactions with K for the FT and WGS reactions.

In this investigation, the FTS reaction was performed using a 1L CSTR on oxide and carburized forms of iron either with or without potassium. For potassium containing catalysts, the Fe_3O_4 phase was obtained following activation with $\text{H}_2/\text{H}_2\text{O}$. FT and WGS rates were compared between oxides and carbides of iron. Iron carbide is found to be active for both FT and WGS reactions. Potassium containing iron catalysts (high α -Fe) yields mainly olefins and oxygenates (aldehyde) after activation with $\text{H}_2/\text{H}_2\text{O}$.

2 Experimental

2.1 Catalyst Preparation

All iron catalysts employed in this study were prepared by following a continuous precipitation method which is described elsewhere [13]. The final compositions of the low α -Fe (less K) and high α -Fe (more K) catalysts used in this study are as follows: 100Fe:4.6Si:2.0Cu:1.4 K (low α -Fe) and 100Fe:4.6Si:2.0Cu:5.0 K (high α -Fe). The above catalysts were calcined in air at 723 K for 6 h. For a comparison purpose, pure Fe_3O_4 was prepared by following the same continuous precipitation technique. A solution of ferric (III) chloride hexahydrate and ferrous (II) sulfate in 1:2 mol ratios was added to water in a round bottom flask. After sealing with a rubber septum, the mixture was purged under nitrogen flow for 2 h and then placed in an oil bath preheated to 353 K. After 10 min, ammonium hydroxide (28 %) was added and the mixture was stirred at 353 K for 1 h. After cooling to room temperature, the mixture was filtered and washed several times using deionized water to remove chloride and sulfate ions adsorbed on the solid. A dark brown precipitate was obtained and the material was dried overnight at 473 K in a static air oven.

2.2 Reaction Testing

The FTS experiment was carried out in a 1 L CSTR. A detailed description of the reactor setup is described elsewhere [13]. Typically, 20.0 g of iron catalyst (140 mesh) was added into melted (413 K) Polywax 3000 (310 g) in the CSTR. Prior to the reaction, the catalyst was activated in situ using $\text{H}_2/\text{H}_2\text{O}$ (1:1) at 513 K and 1.30 MPa for 24 h. The reactor temperature was increased to 543 K and the pressure to 1.30 MPa with a syngas space velocity of 3.0 sl/h/g cat. After 50–100 h time-on-stream (TOS) of FTS, the catalyst was carburized under a CO flow (3.0 sl/h/g cat) at 543 K and 1.30 MPa pressure. The CO and H_2 conversions were obtained by GC analysis of the product gas mixture using a HP Quad series Micro-GC equipped with thermal conductivity detectors. The reaction products

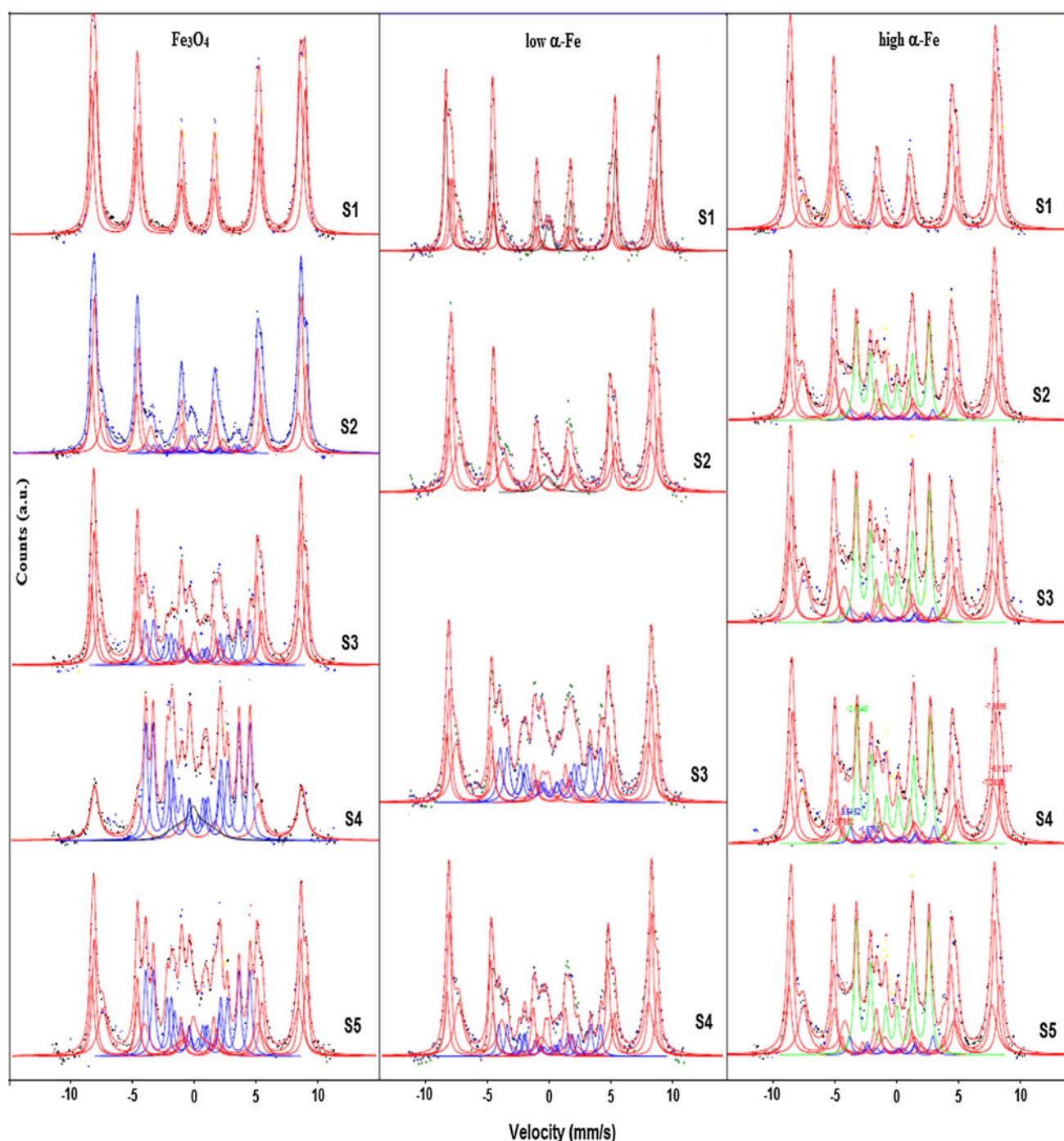


Fig. 2 Mössbauer spectra of Fe_3O_4 (left), low α -Fe (middle) and high α -Fe (right) catalyst samples collected at 20 K after different pretreatment conditions ($\text{H}_2/\text{H}_2\text{O}$ and CO) and during FTS at

543 K, and 1.3 MPa. The fitted curves are shown in solid lines: red, total spectra; red, oxide; blue, χ - Fe_5C_2 ; green, Fe^{2+} . See Tables 1, 2 and 3 for definition of S1, S2, S3, S4 and S5

were collected in three traps maintained at different temperatures: a hot trap (473 K), a warm trap (373 K) and a cold trap (273 K). The products were separated into different fractions (rewax, wax, oil and aqueous) and analyzed separately using a HP 5890 GC with capillary

column DB-5 and a HP 5790 GC with Porapak Q packed column. Catalyst/rewax slurry was withdrawn at different reaction times via a dip-tube after sufficient purging of the collection lines to remove material from the previous sample collection.

Table 1 Summary of phase identification of iron using Mössbauer analysis of fresh and spent unpromoted Fe catalysts

Sample	Conditions	TOS (h)	Fraction of Fe (%) at	
			300 K	20 K
S1	Fresh catalyst (Fe ₃ O ₄)	0	100 in Fe ₃ O ₄ (Fe ₂ O ₃ ?)	100 in Fe ₃ O ₄ (Fe ₂ O ₃ ?)
S2	After WGS (Fe ₃ O ₄)	63	92 in Fe ₃ O ₄ 8 in doublet	91 in Fe ₃ O ₄ 2 in ϵ -Fe ₂ C 5 in χ -Fe ₅ C ₂ 2 in impurities
S3	After FT synthesis (Fe ₃ O ₄)	158.5	67 in Fe ₃ O ₄ 24 in χ -Fe ₅ C ₂ 9 in doublet	69 in Fe ₃ O ₄ 28 in χ -Fe ₅ C ₂ 3 in doublet
S4	After CO activation	182.5	15 in Fe ₃ O ₄ 54 in χ -Fe ₅ C ₂ 31 in doublet	25 in Fe ₃ O ₄ 66 in χ -Fe ₅ C ₂ 9 in impurities
S5	After FT synthesis (carbide)	302.5	48 in Fe ₃ O ₄ 30 in χ -Fe ₅ C ₂ 22 in doublet	67 in Fe ₃ O ₄ 27 in χ -Fe ₅ C ₂ 6 in impurities

2.3 Mössbauer Spectroscopic Measurements

Mössbauer spectra were collected in a transmission mode by a standard constant acceleration spectrometer (MS-1200, Ranger Scientific). A radiation source of 50 mCi ⁵⁷Co in a Rh matrix was used and the spectra were obtained using a gas detector. All samples were investigated at room temperature as well as at cryogenic temperatures (liquid He), typically over a velocity range of ± 10 mm/s. The structural analysis of the samples was done by least-squares fitting of the Mössbauer spectra to a summation of hyperfine sextets. Details about the least-squares fitting procedure are described elsewhere [14].

3 Results and Discussion

The CO conversion values are plotted against TOS for various catalysts after they subject to a different

Table 2 Summary of phase identification of iron using Mössbauer analysis of fresh and spent low α -Fe catalysts

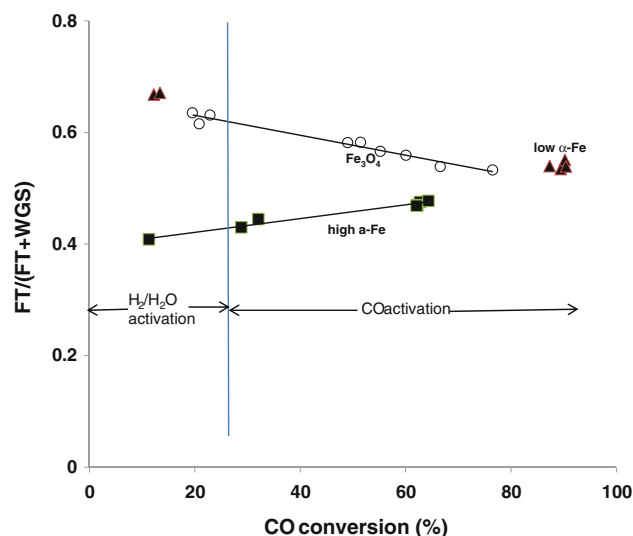
Sample	Conditions	TOS (h)	Fraction of Fe (%) at	
			300 K	20 K
S1	H ₂ /H ₂ O activation	0	30 in Fe ₃ O ₄ 29 in Fe ₂ O ₃ 41 in spm	57 in Fe ₃ O ₄ 40 in Fe ₂ O ₃ 3 in unknown
S2	After FT synthesis (Fe ₃ O ₄)	48	75 in Fe ₃ O ₄ 25 in spm	97 in Fe ₃ O ₄ 3 is unknown
S3	After CO activation	72	45 in Fe ₃ O ₄ 30 in χ -Fe ₅ C ₂ 25 in spm	64 in Fe ₃ O ₄ 33 in χ -Fe ₅ C ₂ 3 in unknown
S4	After FT synthesis (carbide)	120	66 in Fe ₃ O ₄ 18 in χ -Fe ₅ C ₂ 16 in spm	79 in Fe ₃ O ₄ 18 in χ -Fe ₅ C ₂ 3 in unknown

pretreatment conditions (Fig. 1). Independent of the level of potassium loading, the most abundant Fe phase after H₂/H₂O activation is Fe₃O₄. This has been confirmed by Mössbauer analysis of samples obtained after activation (Fig. 2). All three catalysts (Fe₃O₄, low α -Fe, and high α -Fe) have shown some activity after a few hours of TOS for FT synthesis. Among these catalysts, the low α -Fe exhibits higher and steadier CO conversion than the other two catalysts (Fe₃O₄ and high α -Fe) tested after CO activation. The % CO conversion of the Fe₃O₄ catalyst is high initially after CO activation but it deactivated rapidly with TOS. This catalyst lost about 38 % of its initial FT activity in a little over 100 h. On the other hand, the high α -Fe catalyst exhibits a lower CO conversion immediately after the CO pretreatment when compared to the activity following the H₂/H₂O activation. However, the % CO conversion for the high α -Fe catalyst increases with increasing TOS and approaches a steady state value that is higher than after the H₂/H₂O activation. In order to understand this phenomenon, a slurry sample (catalyst+wax) was withdrawn at various stages of the reaction and analyzed for the presence of different Fe phases in an attempt to correlate them with the observed FT activity.

FT synthesis was started initially using all three catalysts (Fe₃O₄, low α -Fe, high α -Fe) which contains mainly Fe₃O₄ (Fig. 2; Tables 1, 2, 3); however, after a period of FT

Table 3 Summary of phase identification of iron using Mössbauer analysis of fresh and spent high α -Fe catalysts

Sample	Conditions	TOS (h)	Fraction of Fe (%) at	
			300 K	20 K
S1	H ₂ /H ₂ O activation	0	83 in sextet 17 in doublet	100 in Fe ₃ O ₄
S2	After FT synthesis (Fe ₃ O ₄)	120	62 in sextet 21 in ϵ' -Fe _{2.2} C 17 in doublet	73 in Fe ₃ O ₄ 20 in ϵ' -Fe _{2.2} C 7 in χ -Fe ₅ C ₂
S3	After CO activation	144	61 in sextet 19 in ϵ' -Fe _{2.2} C 14 in χ -Fe ₅ C ₂ 6 in doublet	69 in Fe ₃ O ₄ 20 in ϵ' -Fe _{2.2} C 11 in χ -Fe ₅ C ₂
S4	After FT synthesis (carbide)	187	61 in sextet 18 in ϵ' -Fe _{2.2} C 14 in χ -Fe ₅ C ₂ 7 in doublet	68 in Fe ₃ O ₄ 21 in ϵ' -Fe _{2.2} C 11 in χ -Fe ₅ C ₂
S5	After FT synthesis (carbide)	307	59 in sextet 22 in ϵ' -Fe _{2.2} C 14 in χ -Fe ₅ C ₂ 5 in doublet	65 in Fe ₃ O ₄ 23 in ϵ' -Fe _{2.2} C 12 in χ -Fe ₅ C ₂

**Fig. 3** Effect of CO conversion on the ratio of FT to WGS rates over unpromoted and Cu, K-promoted Fe catalysts. [CO activation condition: CO (3 sl/h/g cat.), 543 K, 1.3 MPa, 24 h; H₂/H₂O activation condition: 513 K, 1.3 MPa, H₂/H₂O = 1.0, 24 h; FTS condition: 543 K, 1.3 MPa, H₂/CO = 1.0, SV_(syngas) = 3.0 sl/h/g cat.]

synthesis, Fe₃O₄ was partly transformed to carbides, either χ and/or ϵ' depending on the catalyst used. After ~50–100 h of FTS, the Fe₃O₄ catalyst has been converted to a mixture containing 28 % of the Fe in χ -Fe₅C₂, 69 % as Fe₃O₄, and 3 % unidentified (Table 1; Fig. 2). Likewise, the low α -Fe catalyst exhibits 97 % Fe₃O₄ and 3 % unidentified phases (Table 2; Fig. 2) and the high α -Fe catalyst contains 20 % of the Fe as ϵ' -Fe_{2.2}C, 7 % χ -Fe₅C₂ and 73 % Fe₃O₄ (Table 3; Fig. 2). After the initial period, the catalysts were subjected to CO activation at 540 K and 1.3 MPa. This caused a significant increase of the % carbide content in both the Fe₃O₄ (Table 2) and low α -Fe catalysts (Table 3) after the CO activation but such an

Table 4 Activity and selectivity of unpromoted and Cu, K-promoted Fe catalysts under different pretreatment conditions

Catalyst	Activation ^a	CO conv. (%)	Rates (mmol/h/g cat.)		FT/(FT+WGS)	Product selectivity (C-%)				
			FT	WGS		C _n	C _n ⁼	1-OH	1-CHO	Others
Fe ₃ O ₄	None	20.8	4.3	2.7	0.62	55.3	38.0	1.6	0.0	5.1
	CO	55.2	10.4	8.0	0.57	52.7	34.5	1.1	0.0	11.7
Low α -Fe	H ₂ /H ₂ O	20.0	4.2	2.6	0.62	50.3	41.6	1.5	0.0	6.6
	CO	90.2	33.3	27.1	0.55	41.4	43.7	1.3	0.0	13.6
High α -Fe	H ₂ /H ₂ O	22.4	6.0	8.7	0.40	36.6	38.0	6.6	6.0	12.8
	CO	27.0	8.6	9.5	0.48	37.5	41.6	5.9	3.6	11.4
		62.1	10.5	10.3	0.50	36.7	42.7	6.7	1.7	12.2

^a CO activation condition: CO (3 sl/h/g cat.), 543 K, 1.3 MPa, 24 h; H₂/H₂O activation condition: 513 K, 1.3 MPa, H₂/H₂O = 1.0, 24 h; FTS condition: 543 K, 1.3 MPa, H₂/CO = 1.0, SV_(syngas) = 3.0 sl/h/g cat. (C_n paraffin, C_n⁼ olefin, 1-OH 1-alcohol, 1-CHO 1-aldehyde)

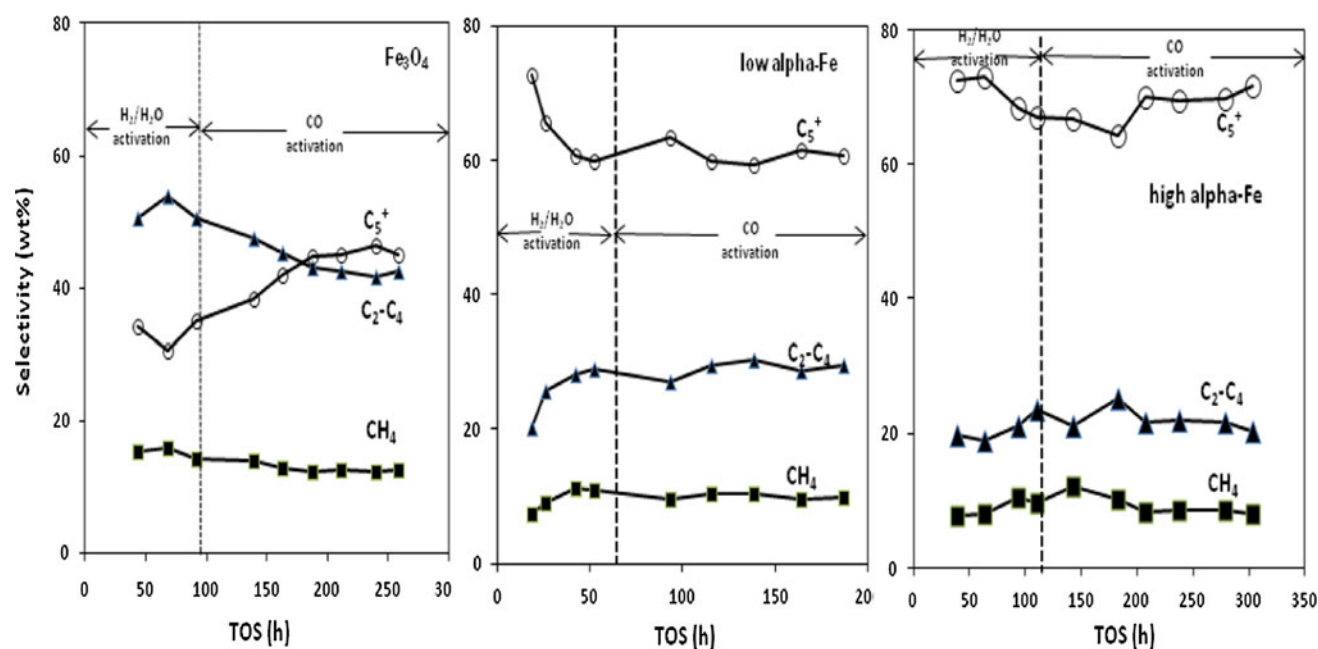


Fig. 4 Reaction time dependence of selectivity over unpromoted and Cu, K-promoted Fe catalysts after different pretreatment conditions. [CO activation condition: CO (3 sl/h/g cat.), 543 K, 1.3 MPa, 24 h;

$\text{H}_2/\text{H}_2\text{O}$ activation condition: 513 K, 1.3 MPa, $\text{H}_2/\text{H}_2\text{O} = 1.0$, 24 h; FTS condition: 543 K, 1.3 MPa, $\text{H}_2/\text{CO} = 1.0$, $\text{SV}_{(\text{syngas})} = 3.0$ sl/h/g cat.]

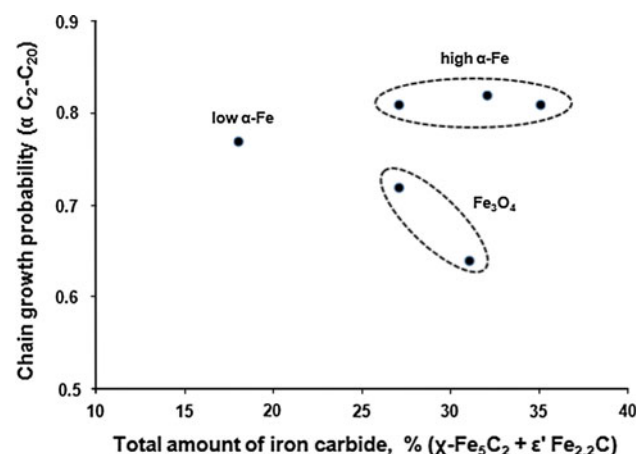


Fig. 5 Chain growth probability (α) versus the total carbide content of various iron catalysts after different pretreatments. [CO activation condition: CO (3 sl/h/g cat.), 543 K, 1.3 MPa, 24 h; $\text{H}_2/\text{H}_2\text{O}$ activation condition: 513 K, 1.3 MPa, $\text{H}_2/\text{H}_2\text{O} = 1.0$, 24 h; FTS condition: 543 K, 1.3 MPa, $\text{H}_2/\text{CO} = 1.0$, $\text{SV}_{(\text{syngas})} = 3.0$ sl/h/g cat.]

increase was not observed in the case of the high $\alpha\text{-Fe}$. In general, iron oxide undergoes a phase transformation from the oxide to carbide under a CO atmosphere at elevated temperatures [4]. Certainly, the rate and the extent of such transition may be determined by many factors such as alkali promoters and other oxides. However, in this case, the high $\alpha\text{-Fe}$ catalyst shows neither an increase in FT activity nor carbide formation after the CO treatment. This

suggests that the presence of a high potassium content in the catalyst led to a carbonaceous layer that covers the iron carbide layer. The high $\alpha\text{-Fe}$ catalyst contained as much carbide following the CO treatment as the low $\alpha\text{-Fe}$ catalyst acquired after about 100 h on-stream.

It is generally agreed that carbides are the most active form of Fe for FT synthesis [15] but the role of Fe_3O_4 for FT and WGS has been under debate for many years. Earlier, Butt et al. [16] suggest that Fe_3O_4 is active for FT synthesis which is in support of the view held by Reymond et al. [17] but in contradiction to the conclusions of Dictor and Bell [18]. Treating Fe_2O_3 with $\text{H}_2/\text{H}_2\text{O}$ allows the reduction of Fe_2O_3 to Fe_3O_4 but not further. As shown in Fig. 1, the three catalysts exhibit low activity when starting as Fe_3O_4 but after treating with CO, the activity of the Fe_3O_4 and low $\alpha\text{-Fe}$ catalysts increased significantly. However, there is a contrast between these two catalysts. The activity of the low $\alpha\text{-Fe}$ catalyst increased initially and then leveled off at about 90 % conversion level. Earlier work with this catalyst composition shows that the activity would decline at about 1 % CO conversion per week. For the unpromoted iron catalyst (Fe_3O_4), the initial activity is similar to the low $\alpha\text{-Fe}$ catalyst but the activity decreases rapidly (ca 30 % per week). The activity of the high $\alpha\text{-Fe}$ catalyst starts low following the activation in CO; however, the CO conversion increases with time and about doubles in one week. Other studies at CAER show that the CO conversion would continue to slowly increase to attain a stable conversion level.

Fig. 6 Effect of catalyst pretreatment conditions on olefinic selectivity over unpromoted and Cu, K-promoted Fe catalysts. (Left, after H_2/H_2O activation; right, after CO activation) [CO activation condition: CO (3 sl/h/g cat.), 543 K, 1.3 MPa, 24 h; H_2/H_2O activation condition: 513 K, 1.3 MPa, $H_2/H_2O = 1.0$, 24 h; FTS condition: 543 K, 1.3 MPa, $H_2/CO = 1.0$, $SV_{(syngas)} = 3.0$ sl/h/g cat.]

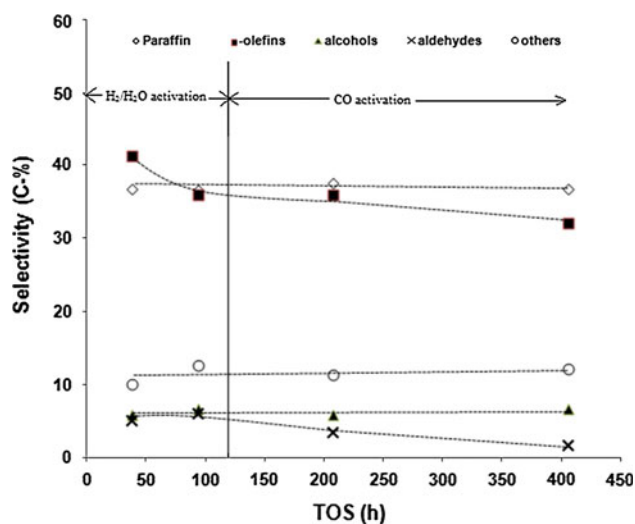
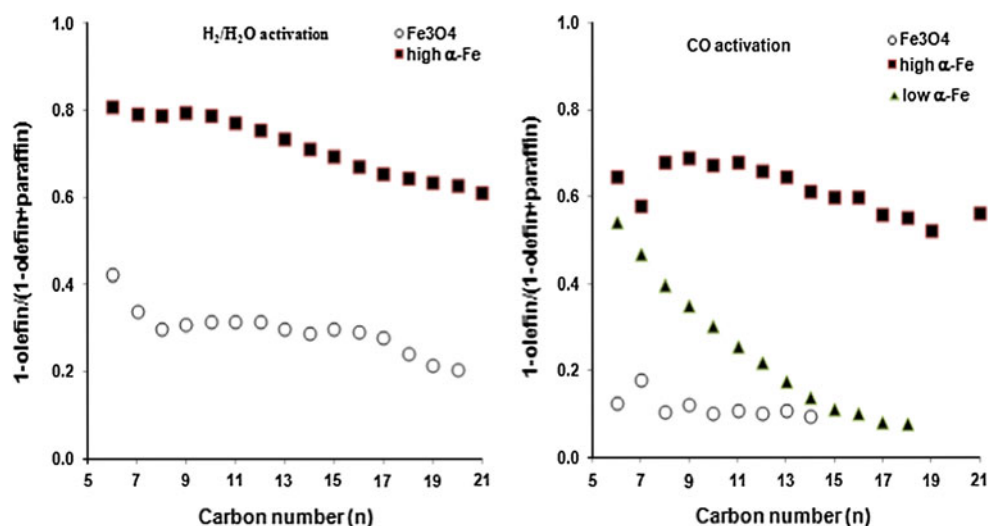


Fig. 7 Reaction time dependence of FTS selectivity over the high- α -Fe catalyst after different pretreatment conditions [FTS condition: 543 K, 1.3 MPa, $H_2/CO = 1.0$, $SV_{(syngas)} = 3.0$ sl/h/g cat.] (others include acids, isomerized alkanes and alkenes, aromatics, etc.)

FT and WGS rates of all the three Fe catalysts were evaluated based on the CO disappearance rate and the rate of CO_2 formation respectively, after pretreatment conditions. As shown in Table 4, the FT and WGS rates for the Fe_3O_4 catalyst increased almost two fold by activating with CO. Similarly, the low α -Fe catalyst exhibits nearly a tenfold increase for FT and WGS activity after CO activation. These increments in rates are compared among the three catalysts using the term $FT/(FT+WGS)$ in Table 4 and Fig. 3. The $FT/(FT+WGS)$ values are about 0.62 for the Fe_3O_4 and low α -Fe catalysts after H_2/H_2O activation and this decreases to 0.57 to 0.55 after CO activation. However, the high α -Fe catalyst initially exhibited a higher WGS activity [$FT/(FT+WGS) = 0.40$] after the H_2/H_2O activation and the FT rate (0.48–0.50) increased with

increasing CO conversion after activating the catalyst in a CO flow for 24 h. From Fig. 3 it is apparent that, irrespective of the catalyst and the iron composition, as the CO conversion increases, the fraction of $FT/(FT+WGS)$ for all three Fe catalysts approaches a value of 0.55. The question of whether the FT and the WGS reaction occur on the same catalytic site is a controversial topic [4, 5, 19–22]. Most reports propose that FT and WGS proceed on different sites, with carbide sites responsible for FT synthesis and Fe_3O_4 as the active phase for the WGS reaction [5]. The present data suggest but does not prove a common site mechanism. It can be concluded from this work that potassium influences both FT and WGS activity by modifying the surface basicity of iron oxides and iron carbides that eventually affect the interactions of H_2 , CO and FT intermediates over the surface of iron. Moreover, the extent with which FT and WGS occur depends on the relative amounts of K present in the catalyst.

The FT product distributions obtained using various Fe catalysts (Fe_3O_4 , low α -Fe and high α -Fe) under H_2/H_2O and CO activation conditions are displayed in Table 4 and Fig. 4. The effect of pretreatment conditions on the selectivity to various hydrocarbons i.e., CH_4 , C_2-C_4 , and C_5+ , are compared for the three Fe catalysts. The selectivity to methane decreased with increasing alkali content, as expected. The selectivity to C_2-C_4 is higher (55, C-%) for the Fe_3O_4 catalyst; as the potassium content increases, the selectivity to C_2-C_4 decreases to 30 and 20 % for low and high α -Fe catalysts, respectively. The C_5+ selectivity of the Fe catalysts was substantially increased with increasing potassium content. However, Fig. 5 shows there is no clear relationship between the chain growth probability and the carbide content of Fe catalysts.

Linear olefins and oxygenates are important classes of products that are obtained from the iron FTS [23, 24]. The influence of reaction conditions on 1-olefin selectivity

Scheme 1 Modified 'CO-insertion' mechanism on alkali-activated iron surfaces, A = Li, Na, K or Cs [35]

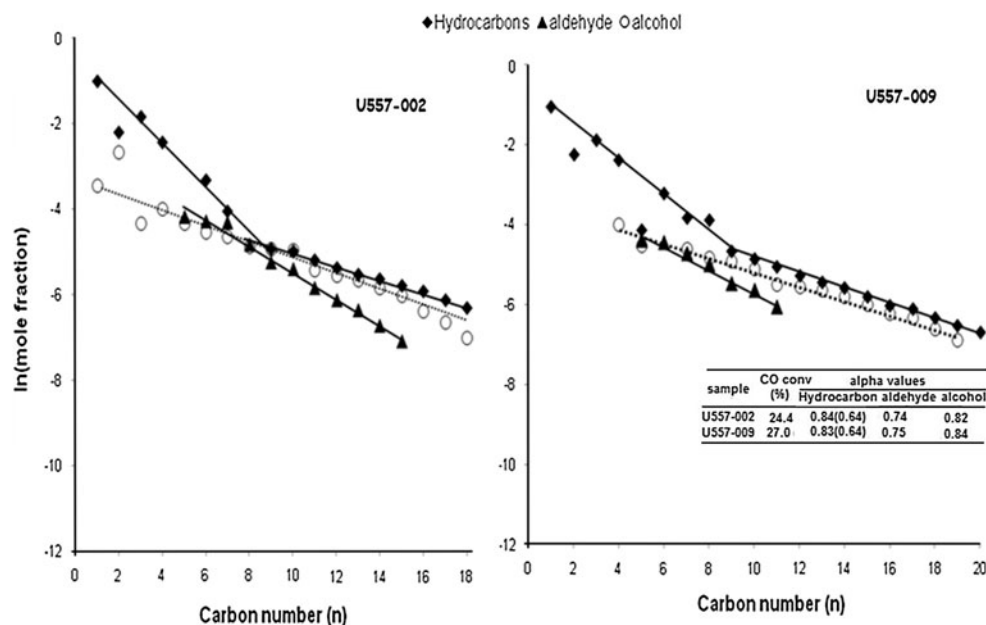
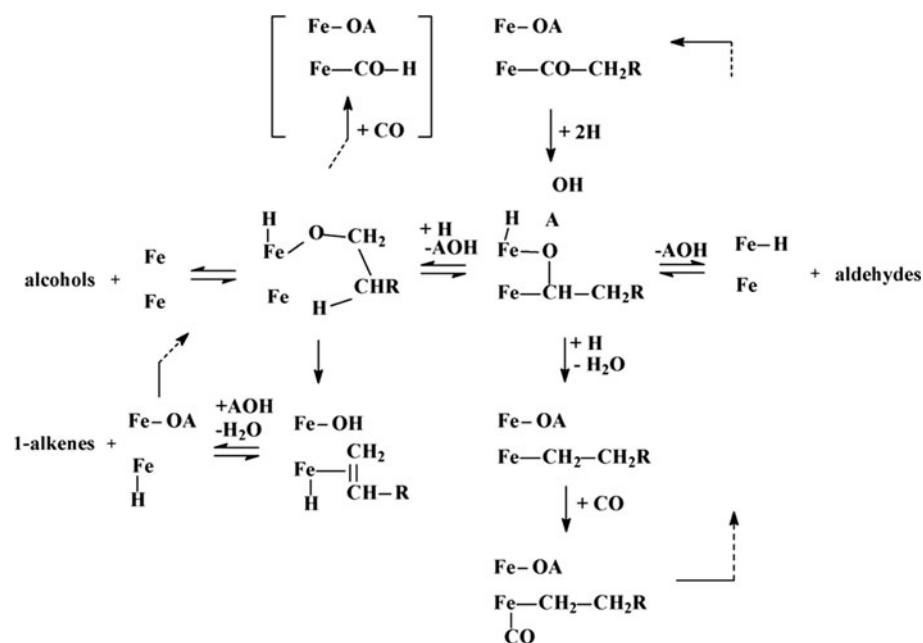


Fig. 8 ASF plot for hydrocarbons, aldehydes and alcohols formed during FTS using high α -Fe catalyst after different pretreatment conditions. ($\text{H}_2/\text{H}_2\text{O}$ activation, U557-002; CO activation, U557-009)

[1-olefin/(1-olefin+paraffin)] for these catalysts are shown in Fig. 6. For a given pretreatment condition, the high α -Fe catalyst yields, as expected, a higher 1-olefin selectivity compared to other two catalysts used in this study. Fe_3O_4 shows a lower 1-olefinic selectivity and its variation with carbon number is shown in Fig. 6. On the other hand, the low α -Fe catalyst exhibits a sharp decline in 1-olefin content with an increase of carbon number following the CO activation. The total olefin content of a particular sample is expressed in terms of % number of moles of carbon found

in the olefins of all carbon numbers compared to the number of moles of carbon for the rest of hydrocarbons, oxygenates and other products which are shown in Table 4. The absence of a significant effect of pretreatment on total olefin content suggests that there is no, or much less, dependency of carbides on the olefin formation. Over all, potassium was found to increase the olefin hydrocarbon content for the Fe-FT synthesis.

The reaction time dependence of FTS selectivities for the high α -Fe catalyst under different pretreatment

conditions are shown in Fig. 7. A significant amount of aldehydes are formed after the $\text{H}_2/\text{H}_2\text{O}$ activation but this dropped from 6.0 to 1.7 (C-%) after the CO activation and increasing TOS; at the same time, the cis-2- and trans-2-olefin content increased after CO activation (not shown).

For Fe-FT synthesis, numerous reaction pathways have been proposed and at least four mechanisms are widely accepted [4, 24, 25]: the so-called ‘carbide’ mechanism [26, 27], alkenyl mechanism [28], enol mechanism [10], and CO-insertion mechanism [29]. However, the ‘CO-inertion’ mechanism is viewed by many researchers as the main reaction pathway leading to the formation of oxygenates [30–32]. Li et al. [33] studied the kinetics for oxygenates formation over Fe–Mn catalysts and suggested that alcohol formation via successive hydrogenation of R-CO_x as intermediate is energetically a more favorable pathway compared to the hydrogenation of acid intermediates. However, our [34] studies have shown that acids, if formed, are readily hydrogenated to the corresponding aldehyde and the alcohol. More recently, Gaube et al. [35] proposed a modified ‘CO-insertion’ mechanism (Scheme 1) in which the alkali participates in the catalytic cycle.

The authors [35] also emphasize the role of acidic Fe-OH groups and further suggest that the polarization of the C–O bond by A^+ facilitates the hydrogenation of the acyl group to 1-hydroxyalkyl or bridging aldehyde intermediates. Claeys et al. [12] reported that potassium might be present as KO instead of K during Fe-FT synthesis. Our EXAFS data show that the Rb alkali in an iron catalyst is present as Rb_2O [14]. Hence, after $\text{H}_2/\text{H}_2\text{O}$ activation, the observed higher amounts of aldehydes and alcohols using the high α -Fe catalyst may be attributed that the present system could follow a modified ‘CO-insertion’ pathway. The data shown in Fig. 8 indicate that there is no significant change in α_2 values for hydrocarbons, aldehydes and alcohols before and after CO activation. However, the total aldehyde content dropped from 6.0 to 1.7 C-% after CO activation and this suggests that K is associated with some form of iron oxides and/or carbides to make the iron carbide surface to have less hydrogenation activity.

4 Conclusion

Iron carbide is found to be active for FT and WGS reactions. Both FT and WGS rates are increased with increasing iron carbide content in the catalyst. Potassium influences the oxides and carbides of Fe differently under FT synthesis conditions. Potassium containing iron catalysts (high α -Fe) yields more olefins and oxygenates (aldehyde) after the $\text{H}_2/\text{H}_2\text{O}$ activation than after formation of iron carbides following a CO treatment. It appears that potassium associated with some iron is responsible for producing oxygenates,

especially aldehyde, during Fe-FT synthesis. A modified ‘CO-insertion’ mechanism may operate for high α -Fe catalyst under the FT reactions conditions used in this study.

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